

Naveed Ahmed

Principal Software Engineer | Technical Lead

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PROFESSIONAL SUMMARY

Principal Software Engineer with 9+ years of experience in software architecture, cloud platforms, and AI/ML enterprise solutions. Proven experience leading end-to-end delivery from domain analysis and data preparation to model training, deployment, and production automation. Extensive hands-on experience using AI coding assistants including Cursor AI and GitHub Copilot to accelerate engineering delivery and improve code quality. Hands-on expertise in C#, .NET, Python, Azure AI Search, Azure AI Foundry, Azure ML, AutoML, MLOps, and OpenAI integrations across healthcare, education, workforce management, and tax/audit domains.

WORK EXPERIENCE

Principal Software Engineer | 10Pearls Pakistan

Jan 2025 - Present | Karachi, Pakistan | Full-time | Hybrid

- Lead technical architecture and engineering strategy for enterprise AI and cloud projects across international clients.
- Architect enterprise RAG platforms, conversational AI systems, and vector search solutions using Azure AI Search, Azure AI services, Pinecone, and OpenAI APIs.
- Owned two tax-domain ML solutions end-to-end by translating legislation into technical requirements, creating synthetic datasets, validating ML approaches, and delivering production-ready outcomes.
- Developed a Python anonymization tool using Azure Language NER and Azure Document Intelligence to anonymize production tax documents for ML training and evaluation.
- Built Azure ML preprocessing pipelines and automated the MLOps lifecycle from preprocessing trigger through managed Azure ML endpoint deployment.
- Used Azure AutoML to run algorithm and hyperparameter searches, selected best-performing models, and deployed them as production endpoints.
- Applied data science techniques to produce data quality reports for model training, identified data issues, and improved feature and label quality.
- Improved performance by up to 5x through architecture optimization and adoption of high-performance retrieval approaches.
- Drive end-to-end delivery using C#/.NET, JavaScript frameworks, CI/CD pipelines, and Infrastructure as Code.
- Set engineering standards, mentor teams, and improve quality through AI-assisted development practices using tools such as Cursor AI and GitHub Copilot.

Technical Lead | Turing.com (Contract)

Jul 2022 - Jan 2025 | Remote, United States

- Led migration of a legacy desktop monolith to a microservices-based SaaS platform with web and mobile applications.
- Implemented APIs, React frontend, React Native app, Azure cloud services, and Kubernetes orchestration.

- Built and mentored development teams, improving onboarding and delivery execution.
- Managed infrastructure, deployments, and project delivery timelines for workforce management solutions.

Senior Software Engineer | VentureDive

Sep 2020 - Apr 2022 | Remote

- Delivered enterprise software for multiple products using .NET, Node.js, TypeScript, React, NestJS, PostgreSQL, and microservices.
- Implemented event-driven integrations with Kafka, Azure Service Bus, AWS SQS, and AWS EventBridge.
- Contributed to architecture, testing, and deployment strategies across Azure and AWS environments.

Senior Software Engineer | Stingray Technologies (Pvt) Ltd

Oct 2017 - Aug 2020 | Karachi, Pakistan | On-site

- Led teams building CPU-intensive and distributed systems using C#, SQL, MongoDB, socket programming, Kafka, and React.js.
- Designed and delivered software solutions with cross-functional stakeholders under tight timelines.

Software Engineer | Techlogix

Dec 2016 - Oct 2017 | Karachi, Pakistan | Full-time

- Developed sales, distribution, and mobility solutions for local and international clients, including Nokia.
- Worked with .NET Web API, MVC, Angular, Bootstrap, Kendo UI, SQL Server, and Oracle.

SKILLS

Programming: C#, .NET Framework, .NET 6/8, JavaScript, TypeScript, Python, Node.js, NestJS, React.js, Next.js, Angular

AI and Machine Learning: OpenAI API (Assistants, Responses, Agents SDK), Azure AI Foundry, Azure AI Search, Azure Machine Learning, Azure AutoML, NLP, Named Entity Recognition (NER), Azure Document Intelligence, RAG, Pinecone, prompt engineering, AI chatbot development, Model Context Protocol (MCP)

Cloud and DevOps: Azure, AWS, Azure ML endpoints, MLOps pipelines, Docker, Kubernetes, CI/CD (Azure Pipelines, Jenkins, GitLab CI), Infrastructure as Code (Bicep, ARM Templates)

Architecture: Microservices, Domain-Driven Design, SaaS multi-tenant architecture, event-driven architecture, SOLID, clean architecture, design patterns

Databases and Storage: SQL Server, MySQL, PostgreSQL, MongoDB, Cosmos DB, Elasticsearch, Redis, Snowflake

Tools and Integrations: Cursor AI, GitHub Copilot, Firebase, Hangfire, Kafka, Azure Service Bus, Google Maps API, Twilio, Square, Shopify

PROJECTS

- **Tax Labs ML Solutions:** Delivered two end-to-end ML solutions by interpreting tax legislation, building synthetic training data, running model experiments, and automating deployment through Azure ML endpoints.
- **Tax Data Anonymization and Preprocessing Pipeline:** Built Python anonymization tooling with Azure Language NER and Azure Document Intelligence, then automated preprocessing and model deployment pipelines in Azure ML workspace.
- **International Audit Firm RAG Application:** Built enterprise RAG solution using Azure AI Search, Azure AI Foundry, Cosmos DB, CI/CD, and Bicep.

- **AI-Powered File Search Optimization:** Designed Pinecone + OpenAI retrieval approach that delivered approximately 5x faster search performance.
- **Conversational Database Interface:** Implemented natural language query workflow using OpenAI and MCP for business-friendly audit insights.
- **Healthcare Education Platform:** Built a full learning platform with exams, question bank, admin portal, website, mobile app, and AI chatbot.
- **Workforce Management SaaS:** Migrated legacy desktop system to microservices with scheduling, payroll, and biometric attendance features.

EDUCATION

Bachelor of Engineering (Computer Engineering)

NED University of Engineering and Technology | 2013 - 2016 | CGPA: 3.86/4.00